Cognitive Computing
UCS420

Landing AI

What is Landing AI ?

Landing AI is a company founded by **Dr. Andrew Ng**, one of the most influential figures in the field of artificial intelligence.

- Landing AI provides a no-code/low-code platform primarily for vision-based AI systems.
- It allows you to train computer vision models, especially for object detection, classification, and defect detection using your own image datasets.
- It focuses on helping businesses—especially those in manufacturing and other traditional industries—**adopt and deploy AI solutions efficiently**.

What is LandingLens ?

1. *“heart”* or *“main tool”* of Landing AI.
2. LandingAI is bringing AI into real-world industries—“The powerful tool it built to make this happen: **LandingLens**.”
3. LandingLens is a no-code computer vision platform.
4. Works even if you don't know how to code.
5. It allows you to train computer vision models, especially for object detection, classification, and defect detection using your own image datasets — perfect for cognitive assistant applications.

“Just like Canva makes design easy for non-designers, LandingLens makes AI easy for non-coders.”

How LandingLens Works ?

Upload images → Label → Train → Deploy

1. Upload images (e.g., product photos from a factory)
2. Label them (mark defects)
3. Train the model (the AI learns patterns)
4. Deploy (use it to automatically detect defects)

Example use case: Defect detection in mobile phone screen

How to Use Landing AI in Experiments

Step 1: Define the Problem (Use Case Selection)

Choose a cognitive assistant task involving vision. Examples:

- Defect detection in mobile phone screen
- Reading labels for medicine bottles.
- Monitoring plant health.
- Smart attendance system using face recognition.
- Detecting emotions from facial expressions.

How to Use Landing AI in Experiments

Step 2: Collect and Prepare Image Dataset

- Use a smartphone or webcam to collect images relevant to the chosen problem. (OR You may upload a dataset (Image Dataset) too)
- Label them based on categories (e.g., "Medicine Bottle", "Laptop", "Book", etc.).
- Ensure data diversity: Different lighting conditions, angles, etc

How to Use Landing AI in Experiments

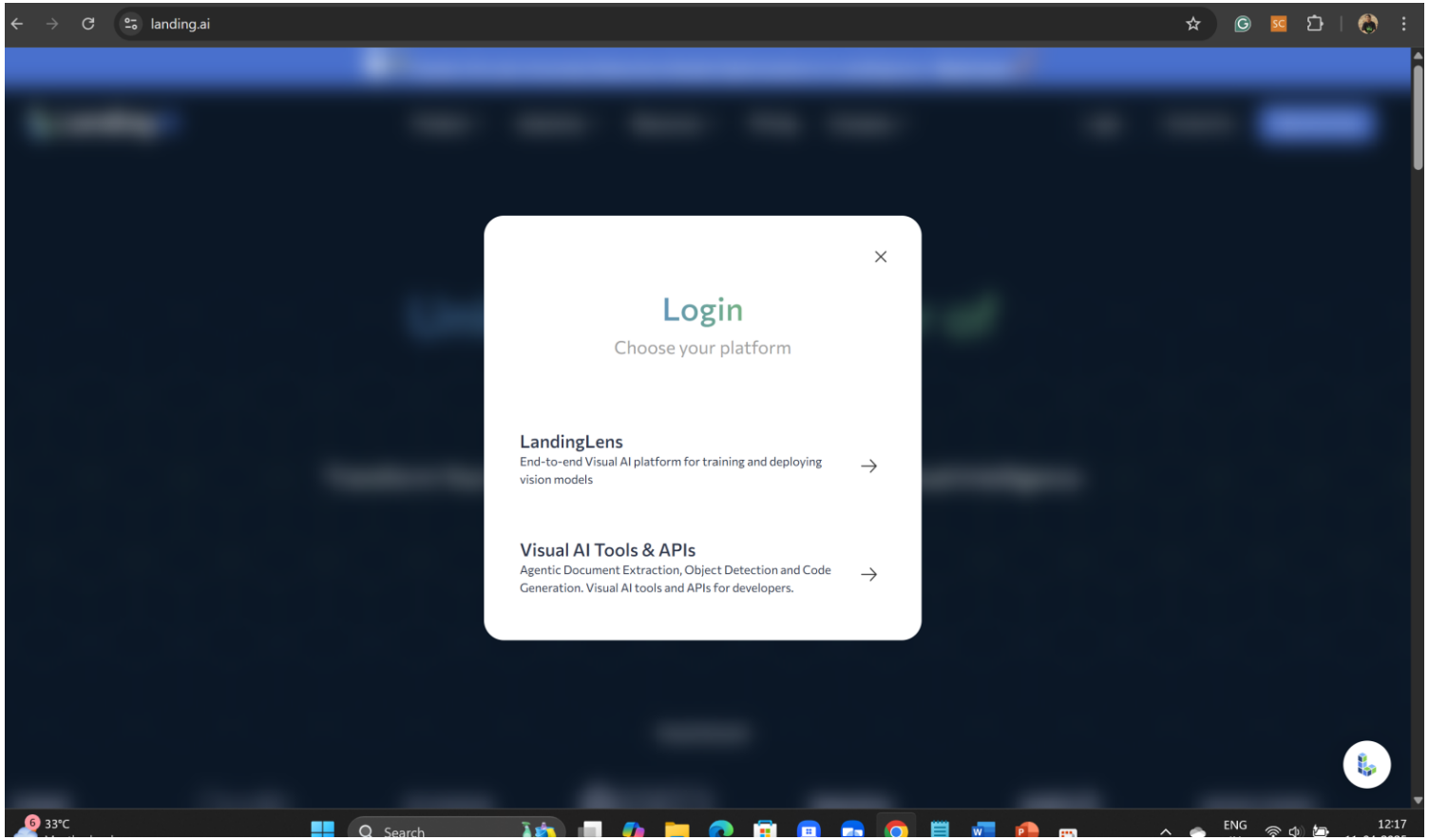
Step 3: Use LandingLens (Landing AI's Platform)

https://www.youtube.com/watch?v=bocvO1eUbVQ&t=327s

1. Create a Project:
   - Log in to https://landing.ai
   - Create a new project (e.g., “Smart Assistant for Visually Impaired”)
2. Upload Images:
   - Upload labeled image datasets.
3. Annotate Data:
   - Use the inbuilt annotation tools (bounding boxes, classification labels).
4. Train the Model:
   - Select the task (classification/detection).
   - Train using the platform’s UI — no coding required.
5. Test the Model:
   - Upload new test images.
   - Evaluate predictions visually and through metrics (accuracy, precision).

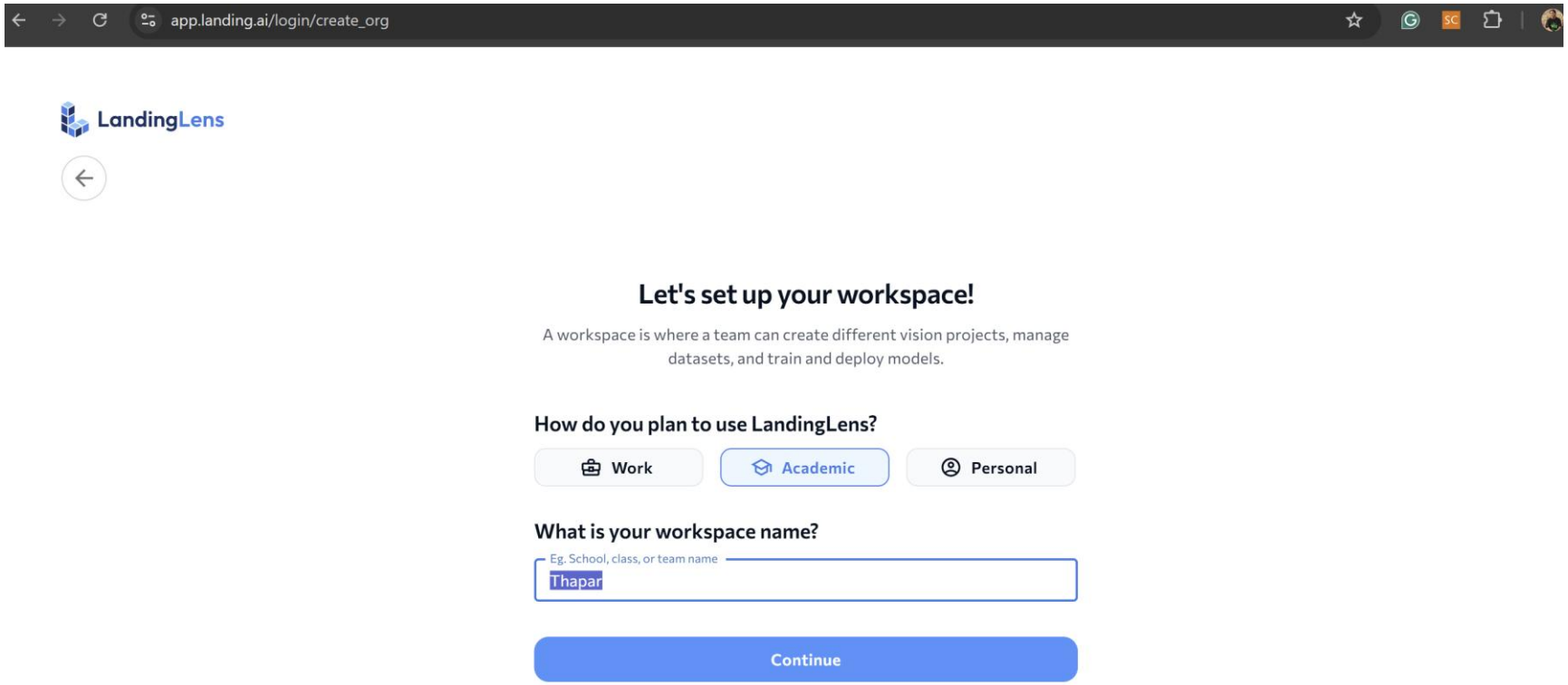
How to Use Landing AI in Experiments Problem-1 : Object Detection

1. Log in to https://landing.ai
2. Then click on LandingLens

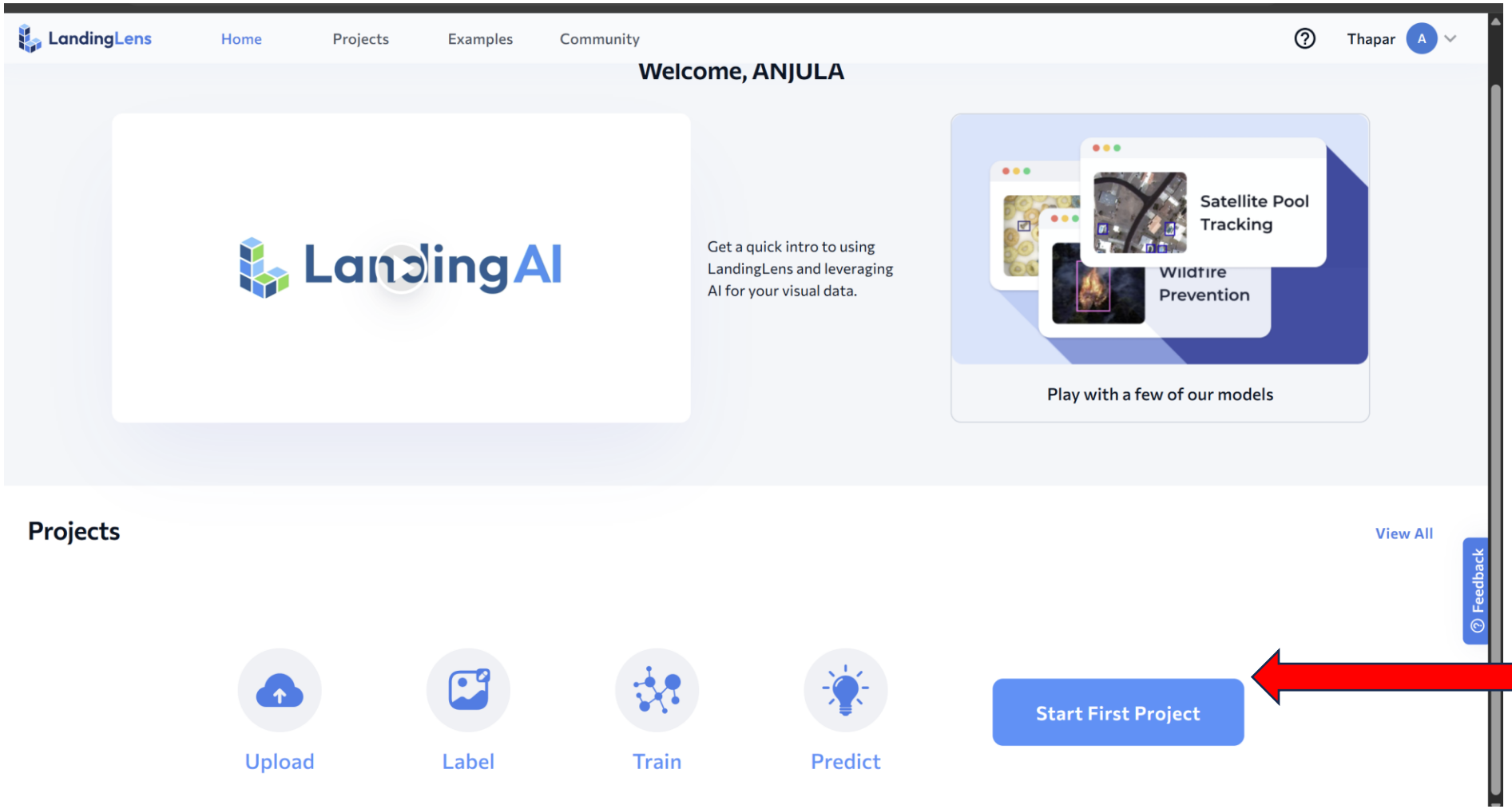


How to Use Landing AI in Experiments Problem-1 : Object Detection

- After login into LandingLens you will see the given below screen:

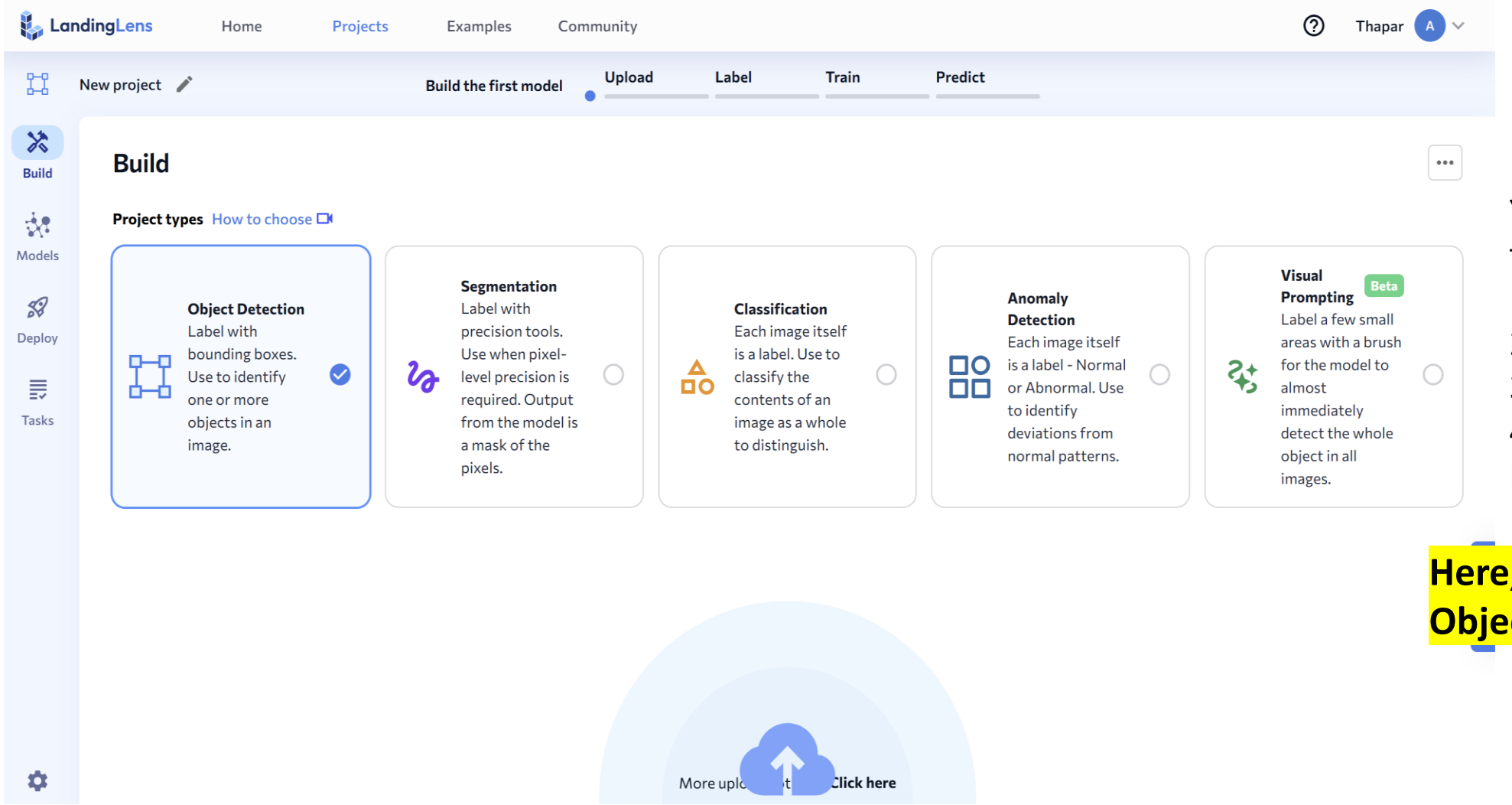


How to Use Landing AI in Experiments
Problem-1 : Object Detection



Click on "Start First Project"

How to Use Landing AI in Experiments
Problem-1 : Object Detection



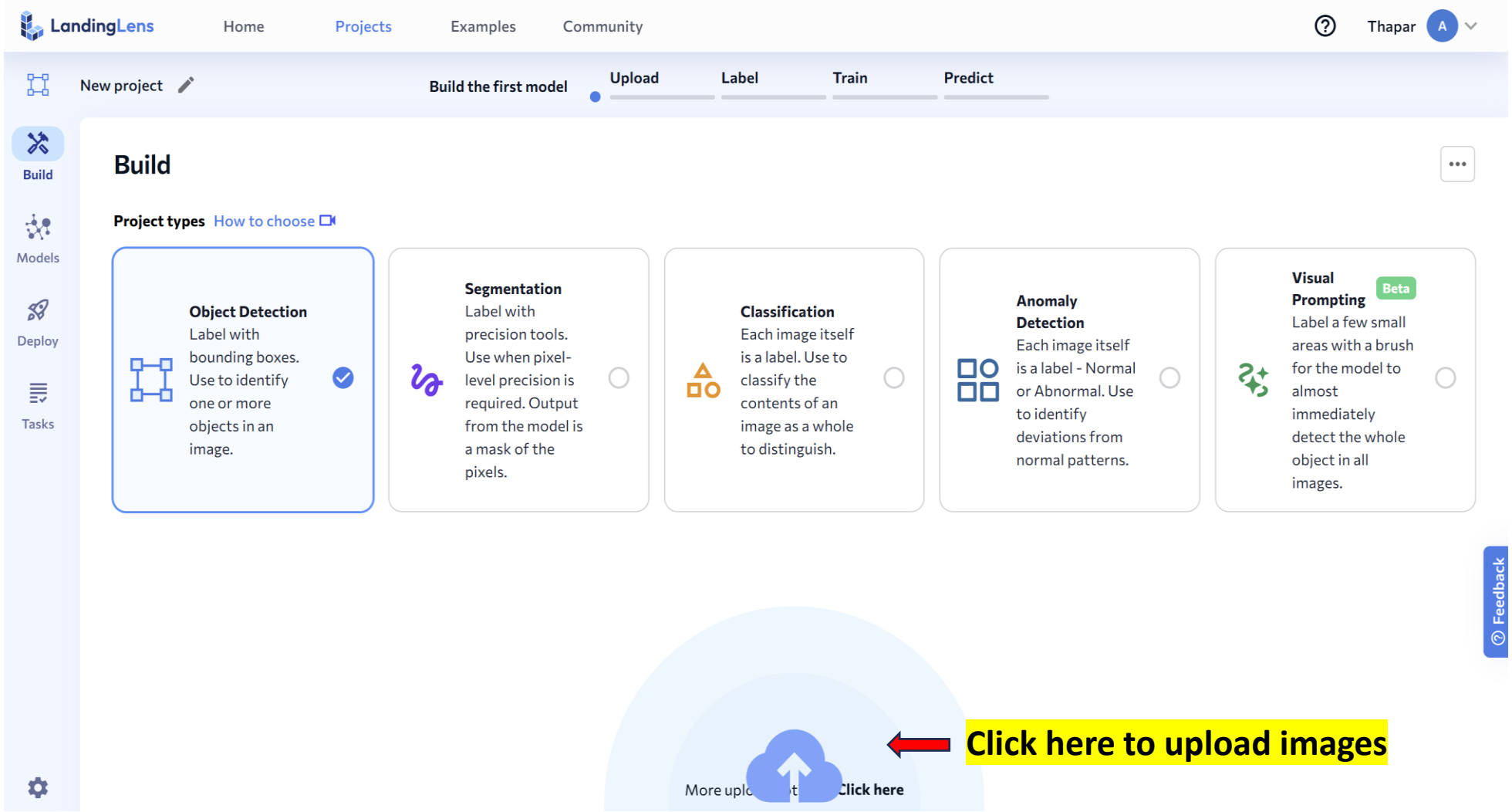
You can create different types of project.

1. Object Detection
2. Segmentation
3. Classification
4. Anomaly Detection

Etc…

Here, we are going to learn Object Detection

How to Use Landing AI in Experiments
Problem-1 : Object Detection



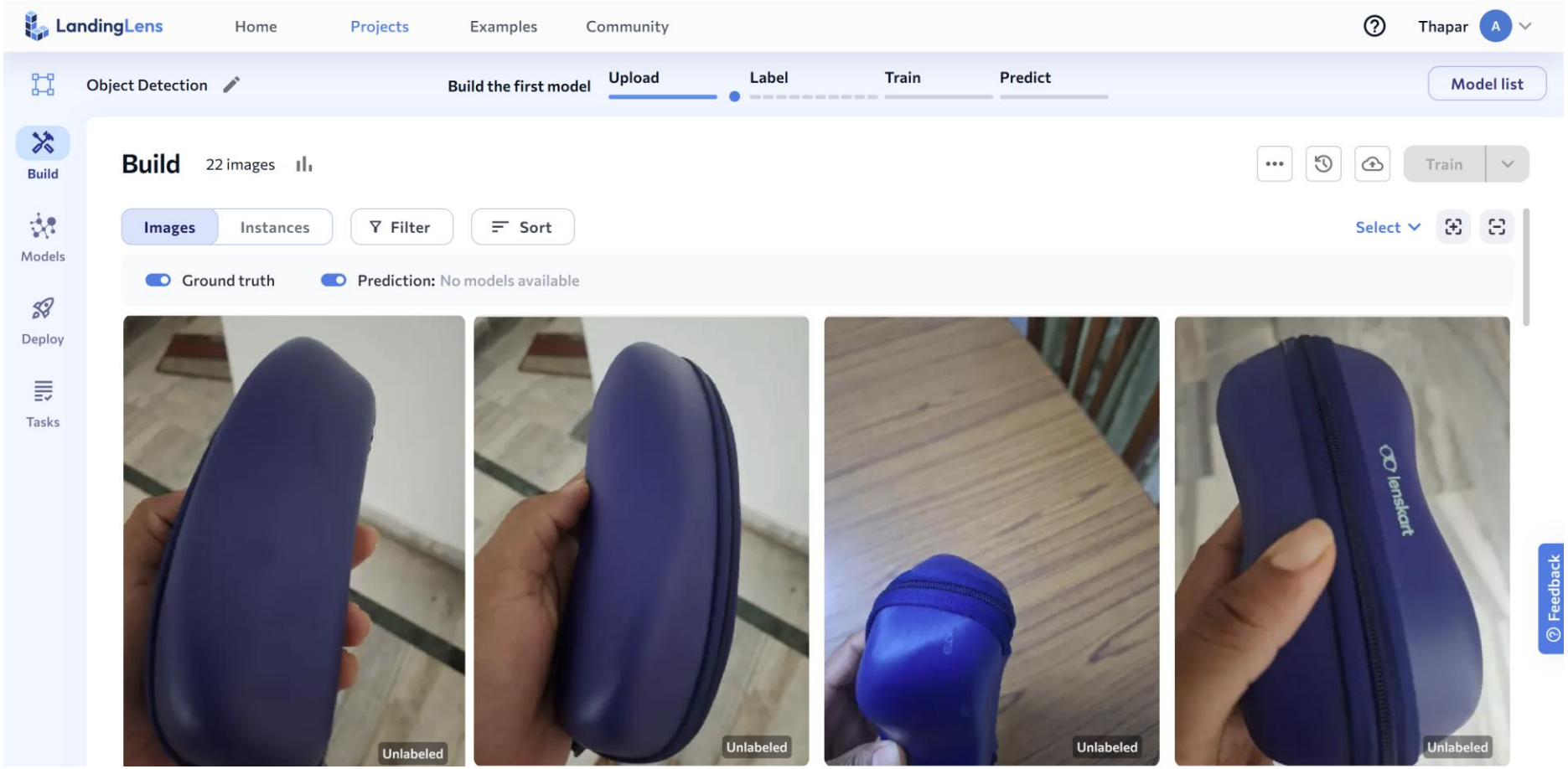
How to Use Landing AI in Experiments Problem-1 : Object Detection



Three options to upload images:

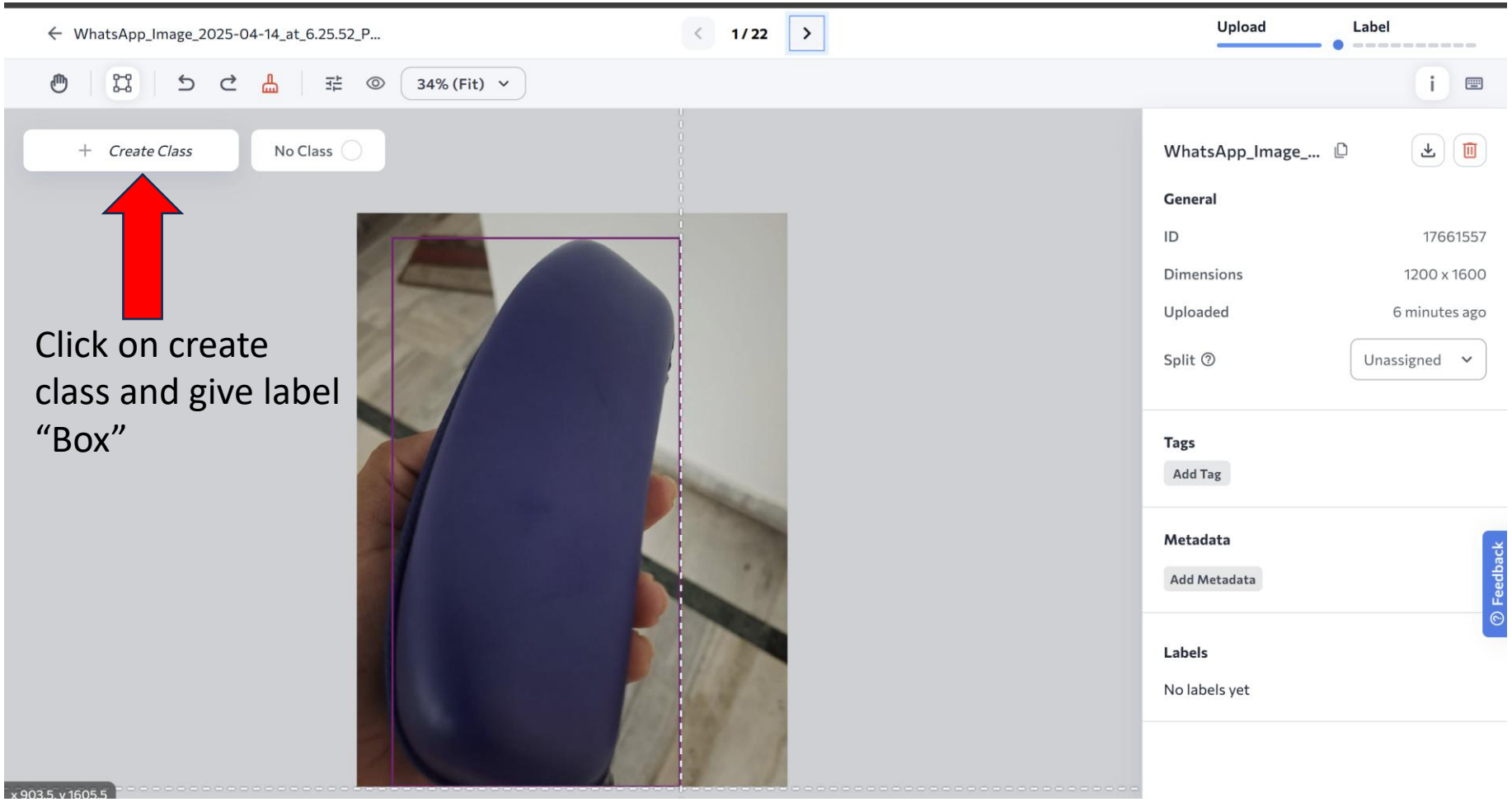
1. **Drag and drop**
2. **Turn on webcam and click pictures.**
3. **Upload from your phone.**

How to Use Landing AI in Experiments Problem-1 : Object Detection

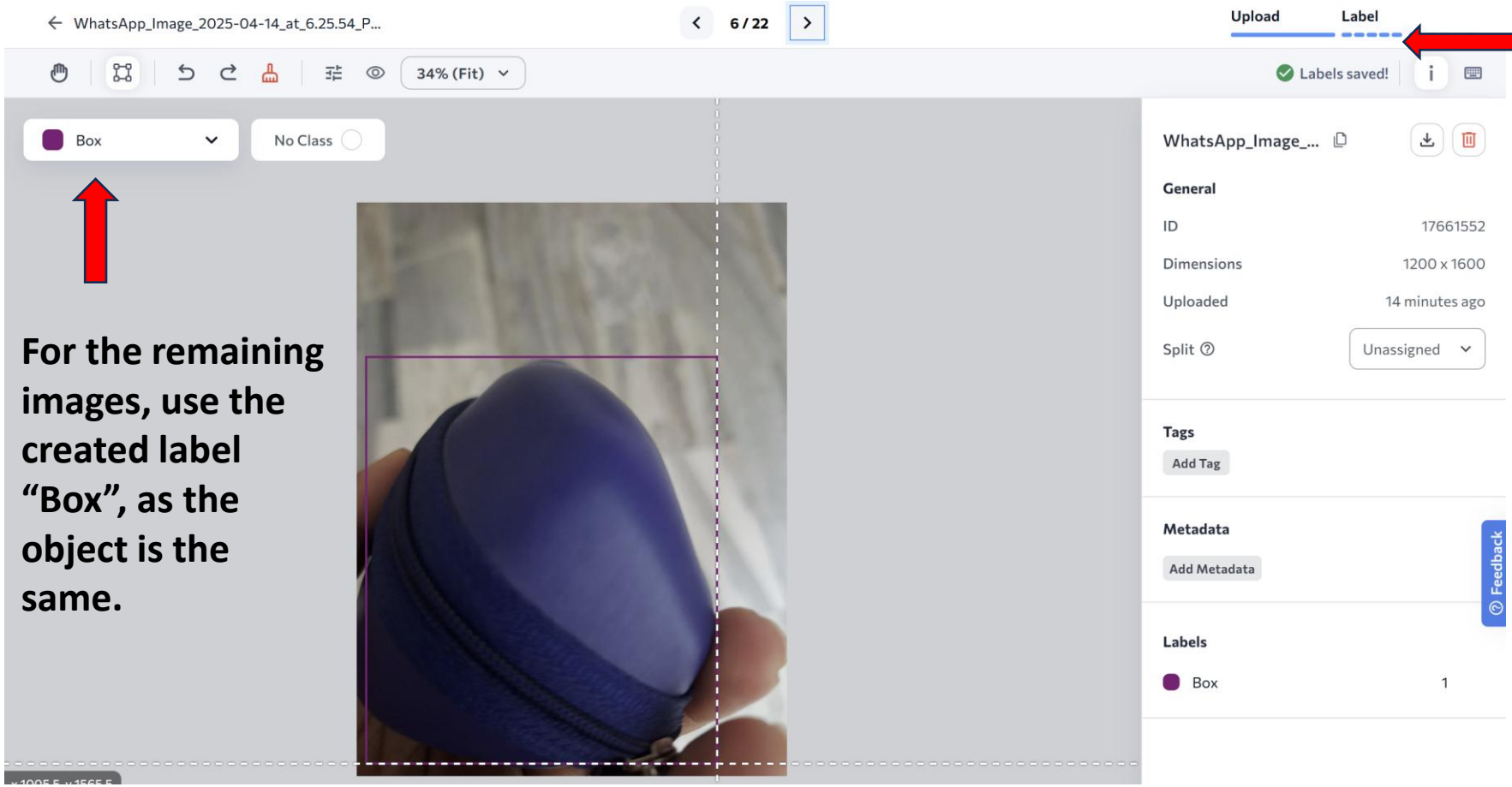


- After uploading the images, next task is to label them (or give class label)
- Click each image one by one and you are allowed to create bounding box. (i.e. give label to each image)

How to Use Landing AI in Experiments Problem-1 : Object Detection



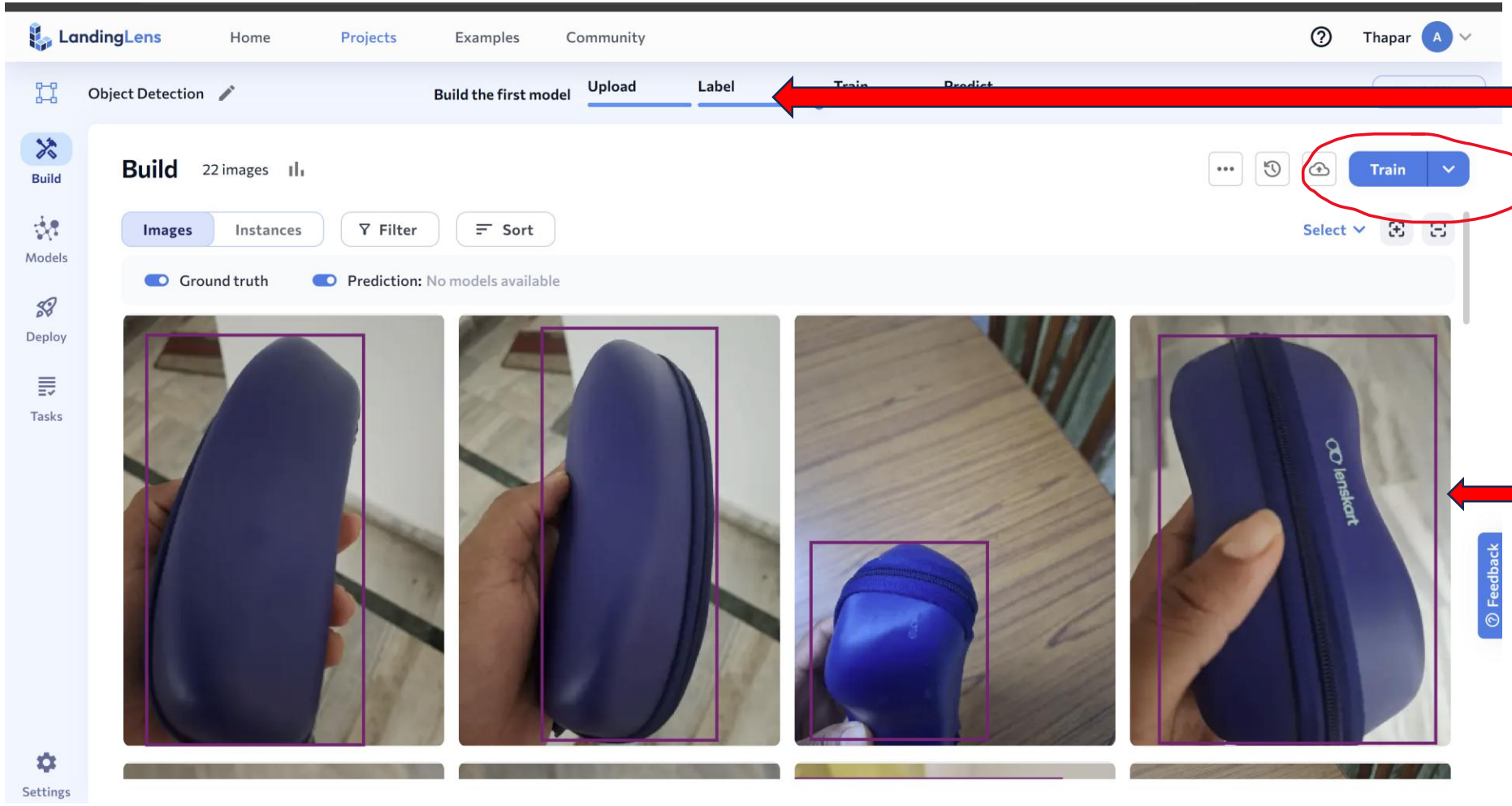
How to Use Landing AI in Experiments Problem-1 : Object Detection



Here you can easily see the total number of images with saved labels.

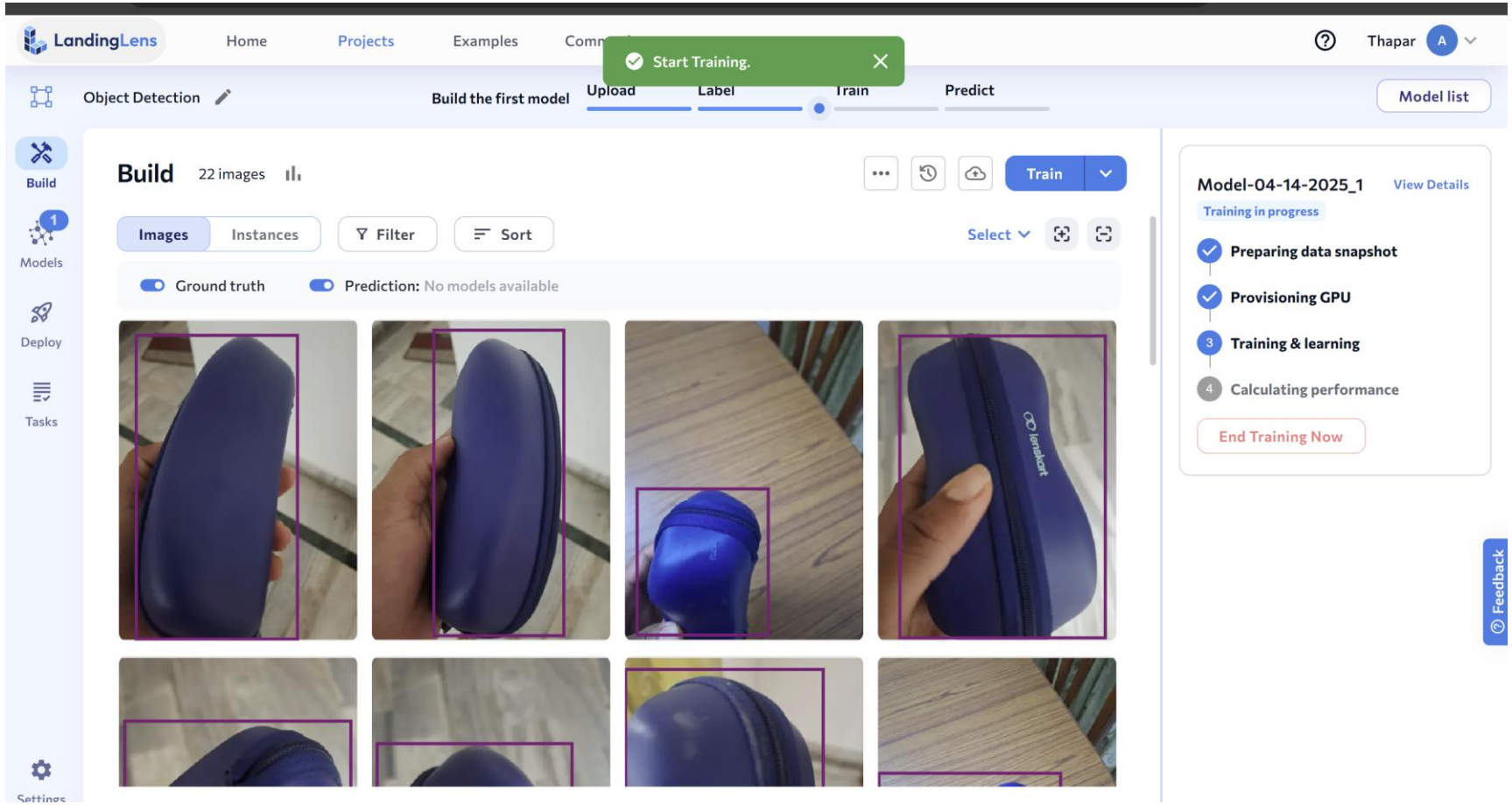
For the remaining images, use the created label "Box", as the object is the same.

How to Use Landing AI in Experiments



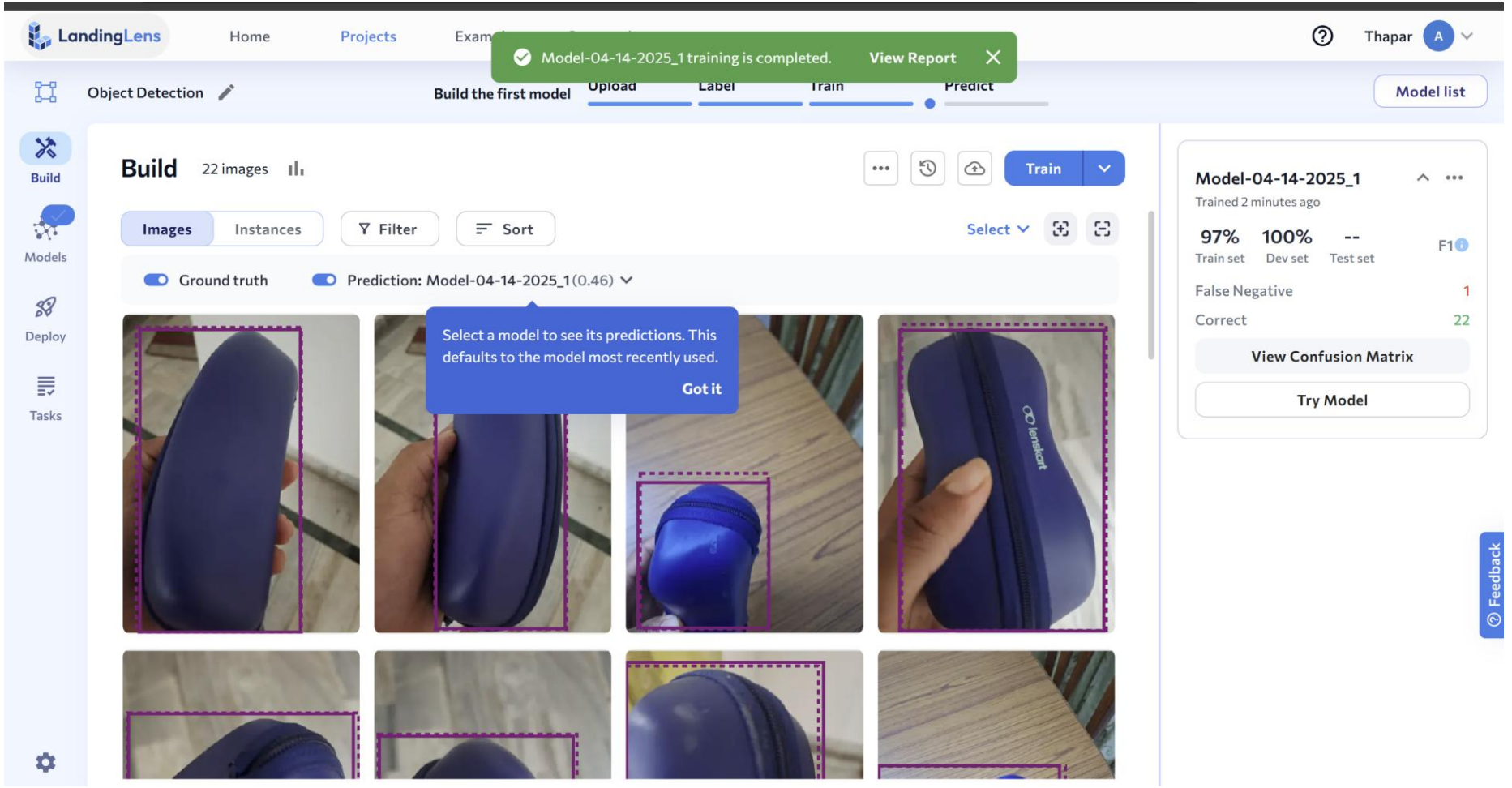
- Labeling is done
- You can easily see the bounding box around each image.“
- Now next task is to train it.
- Click on “Train”

How to Use Landing AI in Experiments



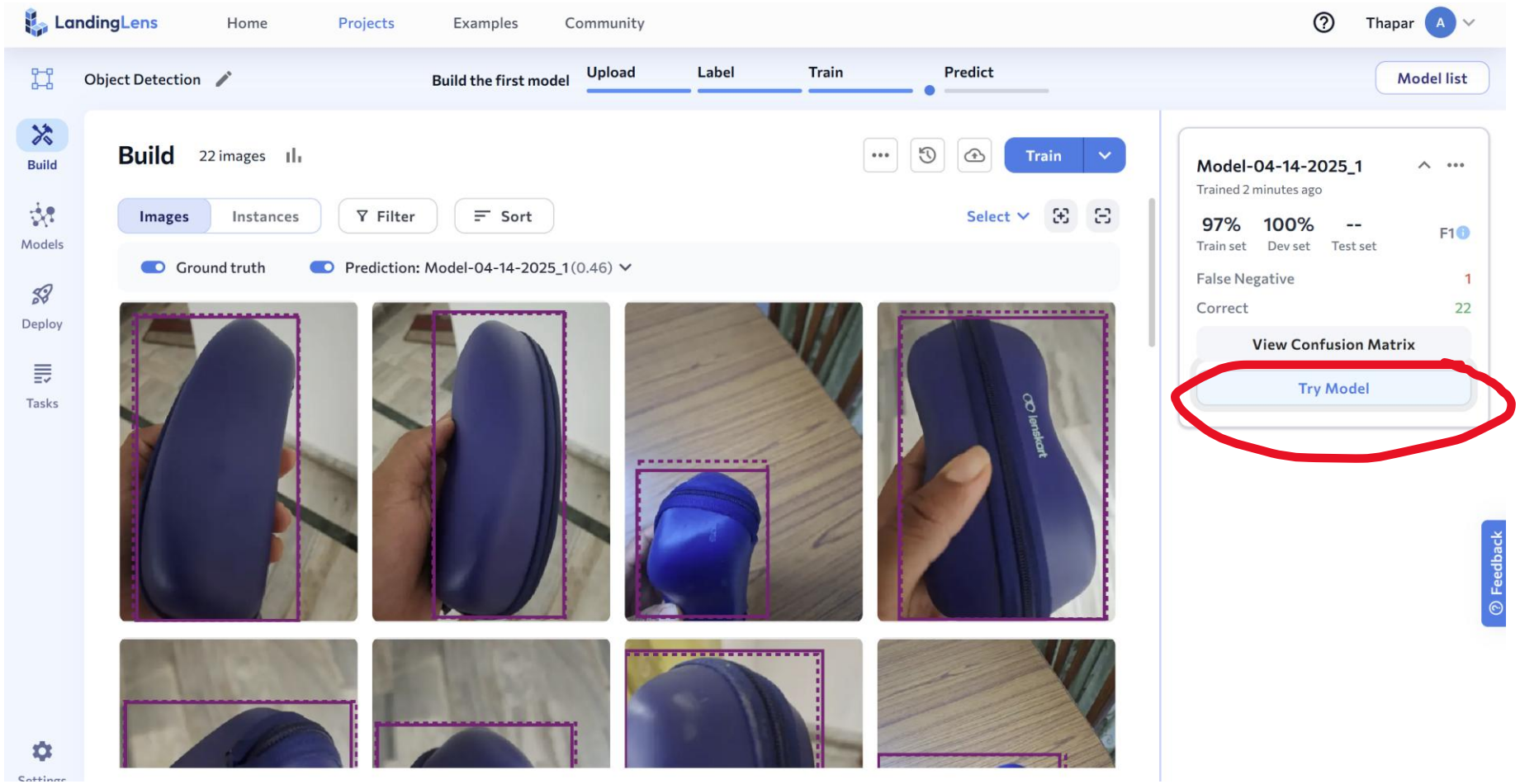
- **Training in progress.**

How to Use Landing AI in Experiments



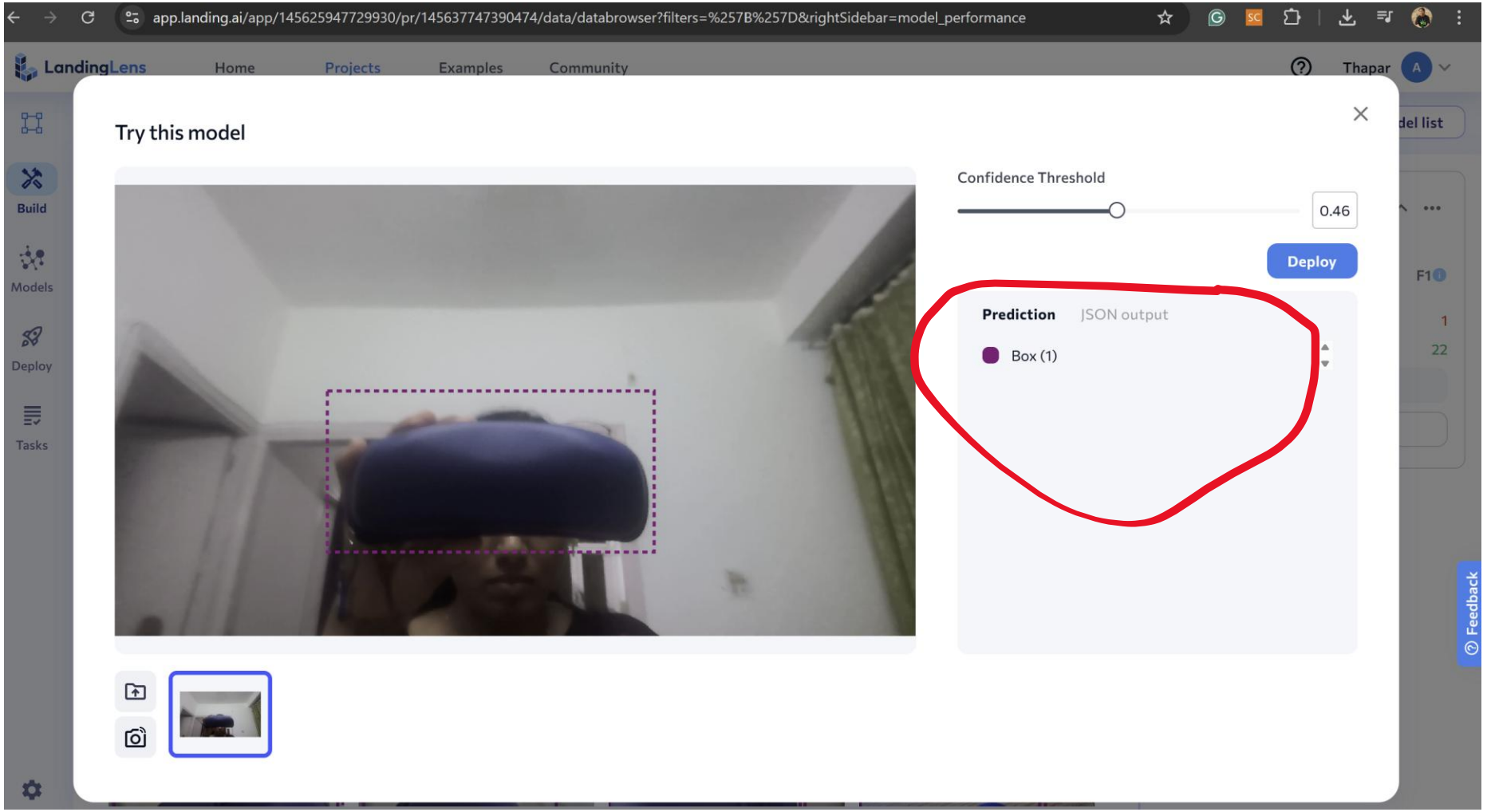
- **Training is completed.**

How to Use Landing AI in Experiments



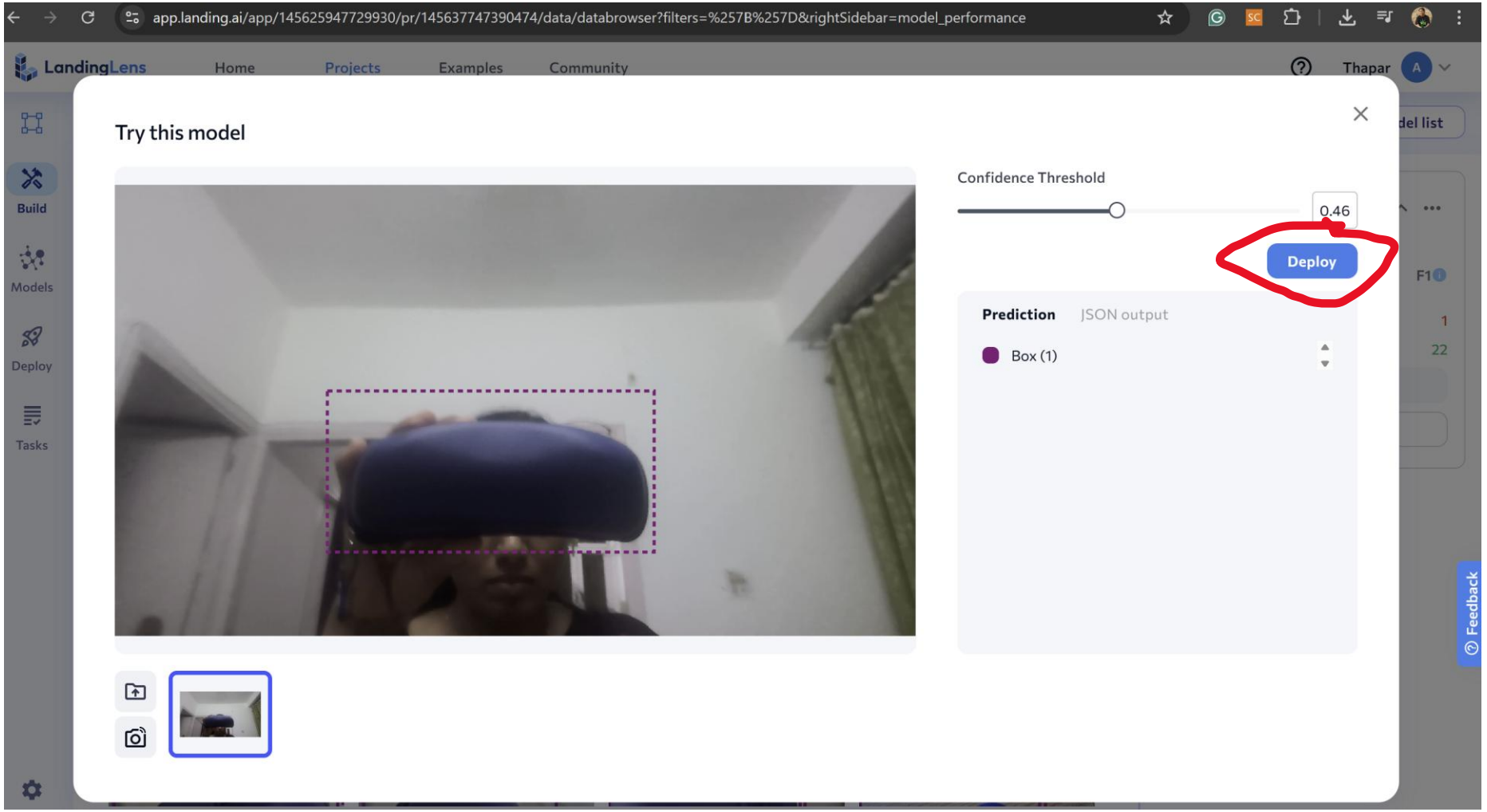
- **Let's Test the Model.**
- **Click on "Try Model"**

How to Use Landing AI in Experiments



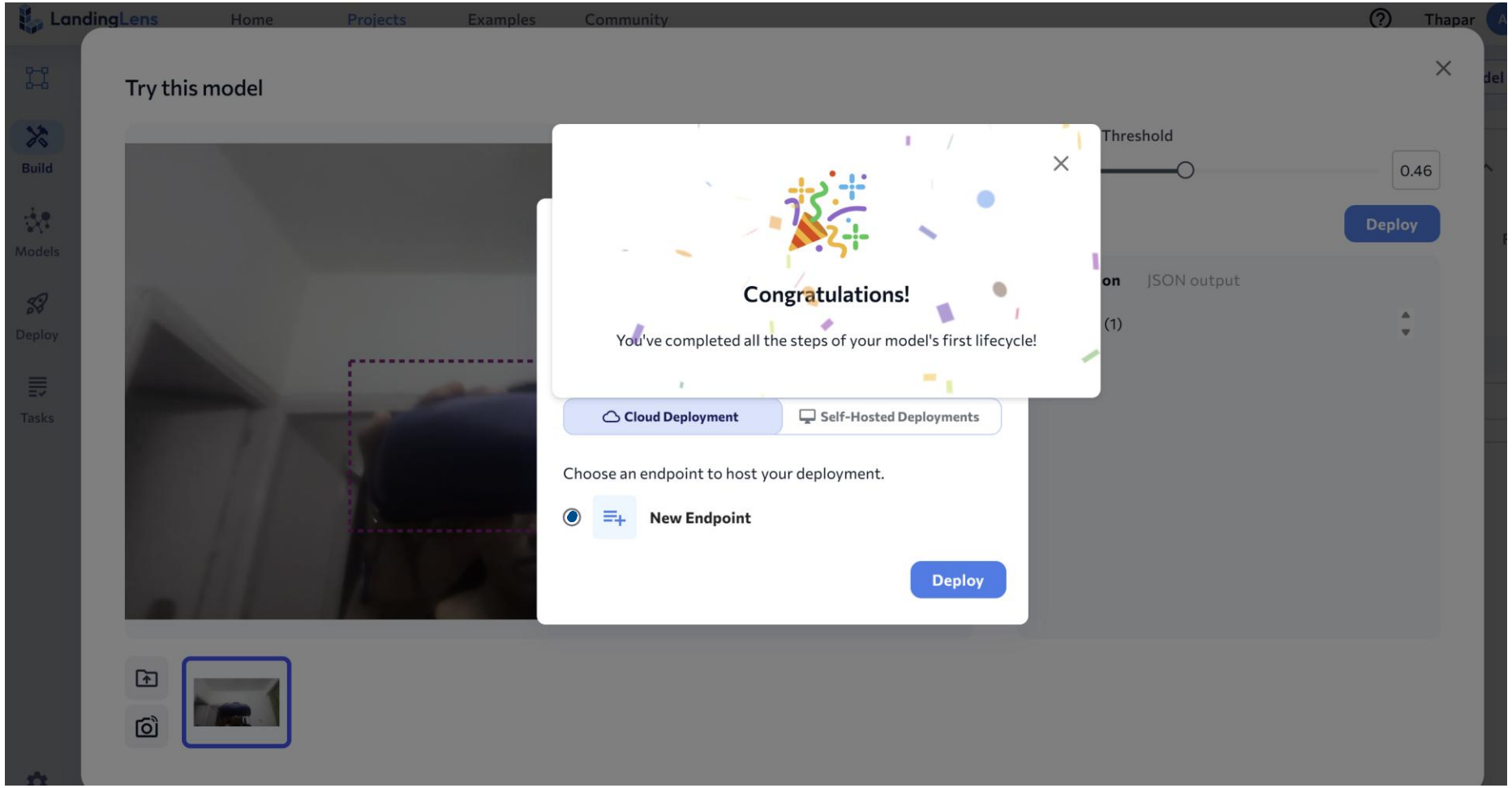
Predicting Correctly

How to Use Landing AI in Experiments



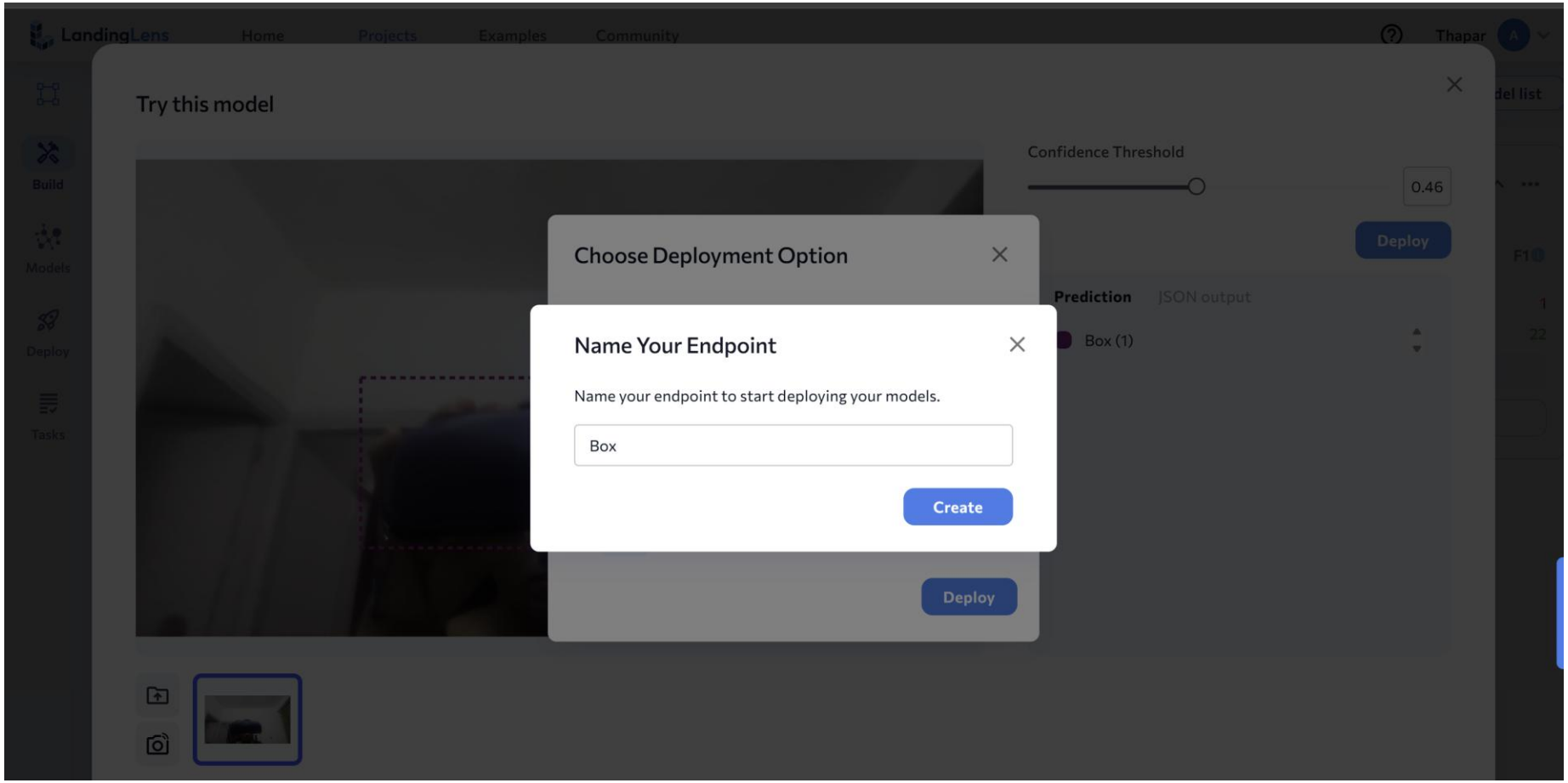
You can deploy it too, i.e. It is going to create a web service.

How to Use Landing AI in Experiments Problem-1 : Object Detection



We will see a web API that it will create.

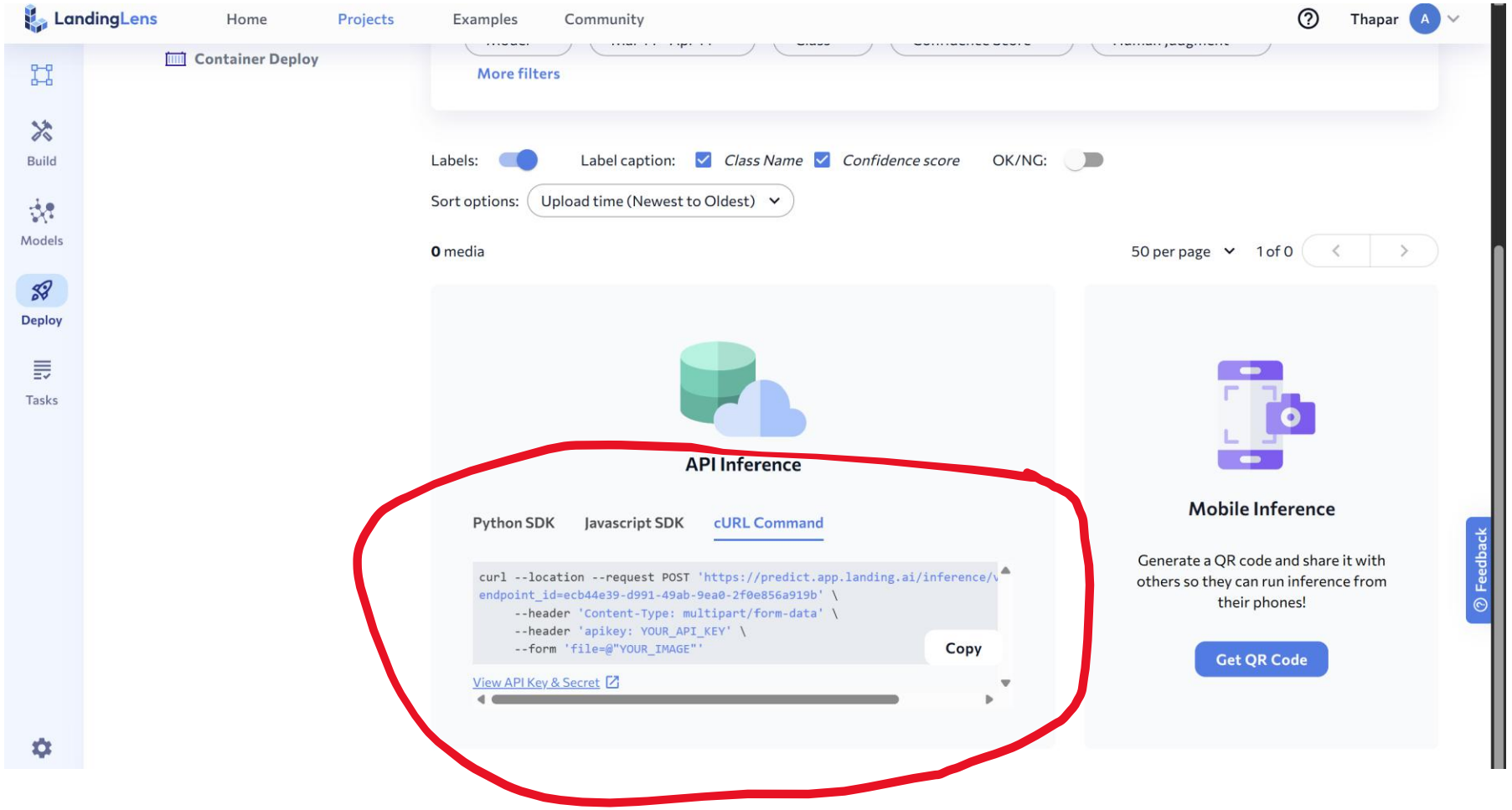
How to Use Landing AI in Experiments Problem-1 : Object Detection



We will see a web API that it will create.

Name the Endpoint, for example- “Box”

How to Use Landing AI in Experiments Problem-1 : Object Detection



- **We will see a web API that it will create.**
- **If you want to deploy this you can simply use these commands.**